

NovaDiff — Change report

Baseline: 9aadd5eb5604

Target: NovaDiff

Generated: 2026-05-27T03:53:47.876Z

Diagrams use Mermaid heuristics (imports, cross-file call guesses, type sketches). They are **not** compiler-grade call graphs. The module graph is file-to-file (JS/TS relative imports). The call graph links files when a callee name maps to exactly one definition in the repo (regex across many languages).

AI project narrative

Workspace overview — roots, comparison intent, scale of change

The baseline snapshot lives in the hidden NovaDiff workspace directory (`/Users/iamgroot/Library/Application Support/novadiff/.../snapshots/9aadd5eb5604`), while the target is the active GitHub repository (`/Users/iamgroot/Documents/GitHub/NovaDiff`).

The diff covers **71 paths: 21 additions** and **50 modifications**; no deletions were reported.

Depth analysis shows most changes at depth 3 (28 files), with 12 at depth 4 and 8 at depth 5.

The change set is concentrated in three top-level segments: `src/` , `packages/` , and `electron/` , indicating a focus on UI, core graph logic, and Electron services.

Change landscape — interpret counts, dominant extensions, depth hotspots, risk intuition

- **File types:** 32 `.ts` , 22 `.tsx` , 8 `.cjs` , 5 `.json` , 3 `.css` , 1 `.rs` .
- **Depth distribution:** 28 files at depth 3, 12 at depth 4, 8 at depth 5.
- **Hotspots:**
 - `electron/` – 7 modified `.cjs` modules (e.g., `code-city-model.cjs` , `git-service.cjs`).
 - `packages/` – 14 modified TypeScript files in `graph-core` .
 - `src/` – 36 modified UI and glue code.
- **Risk signals:**
 - **High-severity:** `cli/src/main.rs` contains patterns `Box::leak(` , `mem::forget(` , `ManuallyDrop::new(` (R1313-R1315).
 - **Medium-severity:** `vite.config.ts` was touched; configuration changes may affect builds.
 - **Medium-high:** `package.json` and `package-lock.json` were modified; dependency versions may shift.
 - **Medium-high:** `electron/git-service.cjs` added parsing helpers (R237-248, R250-269, R271-311, R314-330, R332-33) that could alter Git semantics.

Subsystem map — group paths into coherent areas (config, tests, app, infra, vendor...) using only evidence from the sample paths and top segments

Area	Representative paths	Key changes
Configuration	<code>vite.config.ts</code> , <code>package.json</code> , <code>package-lock.json</code>	Updated build settings and dependencies
Electron services	<code>electron/*.cjs</code> (e.g., <code>code-city-model.cjs</code> , <code>git-service.cjs</code> , <code>knowledge-graph-runner.cjs</code>)	New imports, helper functions, traversal logic
Graph core	<code>packages/graph-core/src/*</code> (analyzer, ignore-filter, tests)	Updated graph builder logic
UI / App	<code>src/*</code> (components, contexts, styles)	UI component updates
CLI tooling	<code>cli/src/main.rs</code>	Rust lifetime patterns, new command logic
Testing	<code>tests/*</code>	Unit tests for layout, git history, security insights
Documentation artifacts	<code>.novadiff-graph/*</code>	Removal of legacy diff overlay and knowledge-graph files

Cross-cutting concerns — security, build/release, migrations, observability (flag unknowns honestly)

- **Security**
 - `electron/code-city-model.cjs` now imports Node's `path` module (line 3); path handling must be validated.
 - New Git parsing helpers in `electron/git-service.cjs` could misinterpret malformed output.
- **Build/Release**
 - Addition of `.cjs` modules and changes to `package.json` may raise the required Node runtime version.

- Rust code in `cli/src/main.rs` references `Box::leak` and `mem::forget` ; build flags and memory safety need review.
- **Migrations**
 - Deletion of `.novadiff-graph/diff-overlay.json` and `.novadiff-graph/knowledge-graph.json` indicates a shift away from the legacy snapshot format.
 - `electron/knowledge-graph-runner.cjs` added `SKIP_DIRS` entries (`"venv"` , `".venv"` , `".venv-main"`) at R32-R34; existing analyses may need re-run.
- **Observability**
 - No new telemetry hooks were added, but the expanded test suite may surface performance regressions in graph layout (`packages/graph-view/src/utils/elk-layout.ts`).
 - Unknown: whether the new CLI memory-leak patterns affect runtime metrics; requires profiling.

Documentation & tooling gaps — what would require Doxygen/clangd/tree-sitter or runtime profiling to validate

- **Rust lifetime safety:** The high-severity patterns in `cli/src/main.rs` cannot be fully verified without a Rust compiler lint run or a memory profiler.
- **Git output parsing:** The helpers added to `electron/git-service.cjs` need integration tests against diverse Git repositories to confirm correctness.
- **Graph traversal correctness:** The updated `SKIP_DIRS` logic in `electron/knowledge-graph-runner.cjs` should be exercised against projects containing virtual environments to ensure they are truly excluded.
- **Electron security:** Runtime checks for path traversal in the renderer should be validated with a sandboxed test harness.

Suggested verification — tests, manual checks, staged rollout

- **Unit tests**
 - Expand tests for `electron/git-service.cjs` to cover edge-case Git status outputs.
 - Add tests for `electron/knowledge-graph-runner.cjs` that simulate projects with nested virtual environments.
- **Integration tests**
 - Run the full graph generation pipeline on a representative set of repositories, comparing output to the previous snapshot format.
 - Verify that the CLI's new memory-leak patterns do not trigger actual leaks by running under `cargo test -- --nocapture` with `valgrind` or `miri`.
- **Manual checks**
 - Inspect the Electron renderer in a sandboxed environment to confirm that the new `path` import does not expose unintended filesystem access.
 - Review the updated `vite.config.ts` for any environment variable changes that could affect production builds.
- **Staged rollout**
 - Deploy the changes to a staging branch and run the full test matrix.
 - Use feature flags to enable the new graph traversal logic in a subset of users before a full release.

Compare metrics

Compare metrics

Metric Count

Added 21

Removed 0

Modified 50

Baseline: 9aadd5eb5604

Target: NovaDiff

Changed paths (sample, up to 80)

- **modified** .novadiff-graph/diff-overlay.json
- **modified** .novadiff-graph/knowledge-graph.json
- **modified** .novadiff-graph/meta.json
- **modified** cli/src/main.rs
- **modified** electron/code-city-model.cjs
- **modified** electron/git-service.cjs
- **modified** electron/knowledge-graph-runner.cjs
- **modified** electron/main.cjs
- **modified** electron/preload.cjs
- **added** electron/security-scan-runner.cjs
- **modified** electron/workspace-history.cjs
- **modified** package-lock.json
- **modified** package.json
- **modified** packages/graph-core/src/analyzer/graph-builder.test.ts
- **modified** packages/graph-core/src/analyzer/graph-builder.ts
- **modified** packages/graph-core/src/ignore-filter.ts
- **modified** packages/graph-view/src/App.tsx
- **modified** packages/graph-view/src/GraphEmbedSyncBridge.tsx
- **modified** packages/graph-view/src/NovaDiffGraphExplorer.tsx
- **modified** packages/graph-view/src/NovaDiffGraphExplorerEmbed.tsx
- **modified** packages/graph-view/src/components/CodeViewer.tsx
- **modified** packages/graph-view/src/components/GraphView.tsx
- **modified** packages/graph-view/src/store.ts
- **modified** packages/graph-view/src/utils/elk-layout.ts
- **added** packages/graph-view/src/utils/elk-layout.worker.ts
- **modified** packages/graph-view/src/utils/louvain.ts
- **added** packages/graph-view/src/utils/performance.ts
- **modified** scripts/run-electron.cjs
- **modified** src/App.css
- **modified** src/App.tsx
- **added** src/app/CodeCityViewportContext.tsx
- **modified** src/app/DocsViewSyncContext.tsx
- **modified** src/app/codeCityChromeInsets.ts
- **modified** src/app/codeCityLayout.ts
- **added** src/app/codeCityRenderer.ts
- **added** src/app/codeMapLoadProgress.ts
- **added** src/app/docsExploreFilters.ts
- **modified** src/app/docsQuality.ts
- **added** src/app/gitHistoryBranches.ts
- **modified** src/app/gitTypes.ts
- **modified** src/app/graphCityBridge.ts
- **added** src/app/graphicsPerformance.ts

- **added** src/app/securityInsights.ts
- **modified** src/app/types.ts
- **modified** src/app/workspaceTypes.ts
- **modified** src/components/CodeCityMinimapSlot.tsx
- **modified** src/components/CodeCityView.tsx
- **added** src/components/CodeMapLoadingPreview.css
- **added** src/components/CodeMapLoadingPreview.tsx
- **modified** src/components/DocsGraphExploreChrome.tsx
- **modified** src/components/DocumentationWorkspace.tsx
- **modified** src/components/DocumentationWorkspaceLinked.tsx
- **modified** src/components/GitHistoryWorkspace.tsx
- **modified** src/components/HistoryCompareStrip.tsx
- **added** src/components/InsightsWorkspace.tsx
- **modified** src/components/KnowledgeGraphPanel.tsx
- **modified** src/components/SidebarNav.tsx
- **modified** src/components/launch/AppLaunchShell.tsx
- **modified** src/components/launch/TypewriterText.tsx
- **modified** src/components/launch/launch.css
- **added** src/components/sidebar/sidebarTiming.ts
- **added** src/components/sidebar/useSidebarExpand.ts
- **added** src/components/sidebar/useWorkspacePageWithSidebar.ts
- **modified** src/vite-env.d.ts
- **added** tests/codeCityLayout.test.ts
- **added** tests/codeMapLoadProgress.test.ts
- **added** tests/docsExploreFilters.test.ts
- **added** tests/gitHistoryBranches.test.ts
- **modified** tests/graphCityBridge.test.ts
- **added** tests/securityInsights.test.ts
- **modified** vite.config.ts

Detected stacks (marker files)

Stack	Markers
node	package.json

Class / symbol index (heuristic)

File	Symbols / defs (regex)
cli/src/main.rs	ChangeKind, path_segments, path_depth, parent_dir, subtree_change_counts, try_build_gitignore_for_root, is_novadiff_docs_reserved_rel, filter_out_novadiff_docs_changes, walk_path_allowed, filter_changes_for_prefetch_gitignore, is_utf8_text_file, is_change_text_eligible, filter_out_non_text_changes, prefetch_summary_queue, empty_summary_evidence, compact_snippet, push_unique_string, line_range_label, detect_file_roles, build_changed_line_ranges, changed_lines_for_side, collect_touched_symbols, is_import_like_line, is_export_like_line, collect_changed_decl_lines, collect_cited_changed_lines, build_verification_hints, build_evidence_badges
electron/code-city-model.cjs	normalizeSide, dominantAuthor, ownershipForRange, selectBlamePaths, buildBlameMap, buildSymbolFileIndex, buildSideNodes, buildCodeCityModelPayload
electron/compare-runner.cjs	cliExecutableName, resolveRustCli, cleanEnv, runCompareEngine, parseEngineStdout, runCompareEngineAsync
electron/diff-excerpt.cjs	buildDiffExcerpt, buildDiffExcerptChunks
electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/file-summary-export.cjs	escapeHtml, escapeMermaidForHtml, injectMermaidFences, markdownToHtml, isReservedNovadiffRel, safeSummarySegments, resolveSummaryDirAbs, safeArtifactSegment, resolveSelectionBaseDirAbs, resolveSelectionDirAbs, buildFileSummaryDocument, buildSelectionSummaryDocument, exportFileSummaryArtifacts, readFileSummaryMarkdowns, exportSelectionSummaryArtifacts, readSelectionSummaryMarkdowns
electron/gh-install.cjs	resolveBrew, getInstallPlan, readGhVersion, getGhToolingStatus, runInstallCommand, installGh
electron/gh-path.cjs	homeDir, extraPathDirs, augmentPathForCli, ghCandidatePaths, resolveViaLoginShell, verifyGh, resolveGhExecutable, clearGhCache, isGhInstalled, ghInstallHint, ghNotFoundError
electron/git-blame.cjs	parseBlamePorcelain, blameFileAtRef, blameFileOwnership
electron/git-commit.cjs	getCommitDetail
electron/git-publish.cjs	buildFullCommitMessage, previewPublish, executePublish
electron/git-service.cjs	gitExecutable, runGit, tryRunGit, isGitRepo, parsePorcelainStatus, getRepoStatus, parseGithubSlugFromUrl, stageAll, stagePaths, unstagePaths, stageDistrict, commit, push, ensureWorktreeBase, removeDirSafe, checkoutRefWorktree, cleanupNovadiffWorktrees, preparePrCompareRoots, splitGitFormatLine, parseLogLines, listBranches, listCommitsForRef, resolveRefHash, detectGitTooling
electron/github-auth-flow.cjs	stripAnsi, cancelGithubDeviceAuth, startGithubDeviceAuth
electron/github-commit-context.cjs	parseJsonSafe, shortGhError, normalizeThread, getGithubCommitContext
electron/github-service.cjs	runGh, tryRunGh, isGhAvailable, getAuthStatus, listRepos, listPullRequests, viewPullRequest, createPullRequest, getUserProfile, listDetectedAccounts, slugFromRepoRoot
electron/graph/extract-structure.mjs	main, buildResult
electron/knowledge-graph-runner.cjs	resolveRepoRoot, resolveNodeExecutable, nodeChildEnv, runCommand, ensureGraphCoreBuilt, loadCore, isTextCandidate, walkProjectFiles, detectLanguage, fileCategory, resolveRelativeImport, buildBatchImportData, runExtractStructureBatch, mapStructuralAnalysis, heuristicComplexity, buildFileMeta, topSegment, buildLayersFromFiles, assignNodeLayers, buildDiffOverlay, getGitHash, buildKnowledgeGraph
electron/llm.cjs	normalizeBase, sendAccumulated, normalizeRepeatUnit, arraysEqual, trimRepeatedParagraphSuffix, trimRepeatedLineSuffix, sanitizeModelResponse, ollamaGenerationOptions, lmStudioGenerationOptions, ollamaNumPredict, streamOllamaChat, lmStudioMaxTokens, streamLmStudioChat, buildWorkspaceDocPrompt, buildCodebaseDocPrompt, buildCommitMessagePrompt, buildSelectionDocPrompt, bulletBlock, renderSummaryEvidenceBlock, buildExplainCodePrompt, buildGroundingContext, verifierWordLimit, buildGroundedRevisionPrompt, shouldRunVerifier, reviseGroundedMarkdown, buildPerFileSummarizePrompt, buildChunkExtractionPrompt, buildChunkSynthesisPrompt

File	Symbols / defs (regex)
electron/main.cjs	sendEngineProgress, getTargetWindow, windowStatePayload, emitWindowState, iconPath, buildMenu, createWindow, sendKnowledgeGraphProgress, sendWorkspaceHistoryProgress, sendGithubAuthProgress
electron/novadiff-docs-html.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, escapeRegex, countMatches, normalizeMalformedCodeFences, looksLikeRawDiffDump, normalizeFenceChunk, normalizeCompactListMarkers, injectMermaidFences, normalizeMarkdownDocument, stripTags, slugify, addHeadingAnchors, markdownToHtml, mermaidSection, navHtml, topDirectory, buildImpactHeatmapHtml, buildSearchIndex, buildEnhancementScript, wrapPage, bundleMetaLine, buildIndexHtml, buildNarrativeHtml, buildMetricsHtml
electron/novadiff-docs-pdf.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, renderSymbolsTable, renderImportEdgesTable, renderDetectedProjects, renderCrossFileCallEdges, mermaidBlock, buildMainPdfHtml, buildDiagramsPdfHtml, waitForPdfReady, htmlToPdfFile, generateNovadiffDocsPdf
electron/novadiff-docs-writer.cjs	bundleKeyOf, bundleBaseDir, bundleHeadline, readme, writeNovadiffDocsBundle
electron/prefetch-summaries.cjs	clearSummaryPrefetchCache, stopSummaryPrefetchWorker, getPrefetchedSummary, defaultPrefetchLimit, runSummaryPrefetchLoop, startSummaryPrefetchWorker
electron/repo-discovery.cjs	homeSearchRoots, readGitRemoteSlug, scanForGitRepos, discoverRepos, matchRepoForGithubRepo
electron/security-scan-runner.cjs	severityFromAdvisory, confidenceFromSeverity, hashId, relPathFromRoot, walkRepoForMarkers, detectEcosystems, runCommand, parseNpmAudit, parsePipAudit, parseCargoAudit, parseOsvScanner, runSecurityInsightScan
electron/workspace-files.cjs	listSnapshotFiles, readSnapshotTextFile
electron/workspace-history.cjs	removeDirSafe, isUsableSnapshotDir, listCommitsOldestFirst, snapshotCommit, ensureCommitSnapshot, indexWorkspaceHistory, refreshWorkspaceHistory, materializeCommitForCompare
electron/workspace-store.cjs	sessionPath, workspacesRoot, readJson, withSessionWriteLock, writeJsonAtomic, writeJson, loadSession, setLocalOnlyMode, saveSession, listWorkspaces, upsertWorkspace, setActiveWorkspace, setGitUser, getWorkspace, updateWorkspaceUiState, updateWorkspaceLiveRepo, createWorkspace
packages/graph-core/scripts/generate-alphabetical-catalog.mjs	normalizeKey, slugify, resolveWasmsKey, resolveExtensions, claimExtensions
packages/graph-core/src/__tests__/change-classifier.test.ts	makeAnalysis
packages/graph-core/src/__tests__/plugin-registry.test.ts	createMockPlugin
packages/graph-core/src/__tests__/staleness.test.ts	makeGraph
packages/graph-core/src/analyzer/graph-builder.ts	FileMeta, FileAnalysisMeta, NonCodeFileMeta, NonCodeFileAnalysisMeta, GraphBuilder
packages/graph-core/src/analyzer/language-lesson.ts	LanguageLessonResult, buildConceptPatterns, detectLanguageConcepts, getLanguageDisplayName, buildLanguageLessonPrompt, extractJson, parseLanguageLessonResponse
packages/graph-core/src/analyzer/layer-detector.ts	LLMLayerResponse, toLayerId, matchFileToLayer, detectLayers, buildLayerDetectionPrompt, parseLayerDetectionResponse, applyLLMLayers
packages/graph-core/src/analyzer/llm-analyzer.ts	LLMFileAnalysis, LLMProjectSummary, buildFileAnalysisPrompt, buildProjectSummaryPrompt, extractJson, parseFileAnalysisResponse, parseProjectSummaryResponse
packages/graph-core/src/analyzer/normalize-graph.ts	stripToValidPrefix, normalizeNodeId, normalizeComplexity, DroppedEdge, NormalizationStats, NormalizeBatchResult, inferTypeFromId, normalizeBatchOutput
packages/graph-core/src/analyzer/tour-generator.ts	buildTourGenerationPrompt, parseTourGenerationResponse, generateHeuristicTour
packages/graph-core/src/change-classifier.ts	UpdateDecision, classifyUpdate, detectDirectoryChanges, topDirectory, summarizeChanges
packages/graph-core/src/embedding-search.ts	SemanticSearchOptions, cosineSimilarity, SemanticSearchEngine
packages/graph-core/src/fingerprint.ts	FunctionFingerprint, ClassFingerprint, ImportFingerprint, FileFingerprint, FingerprintStore, FileChangeResult, ChangeAnalysis, contentHash,

File	Symbols / defs (regex)
	extractFileFingerprint, compareFingerprints, buildFingerprintStore, analyzeChanges
packages/graph-core/src/ignore-filter.ts	IgnoreFilter, createIgnoreFilter
packages/graph-core/src/ignore-generator.ts	parseGitIgnorePatterns, isCoveredByDefaults, generateStarterIgnoreFile
packages/graph-core/src/languages/catalog-builder.ts	catalogEntryToLanguageConfig, buildAlphabeticalLanguageConfigs, listAlphabeticalLanguages
packages/graph-core/src/languages/catalog-types.ts	LanguageCatalogEntry
packages/graph-core/src/languages/framework-registry.ts	FrameworkRegistry
packages/graph-core/src/languages/language-registry.ts	LanguageRegistry
packages/graph-core/src/languages/merge-configs.ts	mergeLanguageConfigs, languageConfigsForRegistry
packages/graph-core/src/persistence/index.ts	ensureDir, sanitiseFilePaths, saveGraph, loadGraph, saveMeta, loadMeta, saveFingerprints, loadFingerprints, saveConfig, loadConfig, saveDomainGraph, loadDomainGraph
packages/graph-core/src/plugins/discovery.ts	PluginEntry, PluginConfig, parsePluginConfig, serializePluginConfig
packages/graph-core/src/plugins/extractors/__tests__/cpp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/csharp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/go-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/java-extractor.test.ts	parse, Repository
packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	parse, helper, greet, noReturn, noop, multiline, Loggable, Repository, publicFunc, anotherFunc, myFunc, process, main, helper, findUser
packages/graph-core/src/plugins/extractors/__tests__/python-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/ruby-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/rust-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/base-extractor.ts	traverse, getStringValue, findChild, findChildren, hasChildOfType
packages/graph-core/src/plugins/extractors/cpp-extractor.ts	unwrapDeclaratorName, extractFuncDeclName, extractParams, extractReturnType, isStatic, CppExtractor
packages/graph-core/src/plugins/extractors/csharp-extractor.ts	extractParams, extractReturnType, hasModifier, extractUsingSource, lastComponent, extractInvocationName, CSharpExtractor
packages/graph-core/src/plugins/extractors/generic-extractor.ts	nodeName, lineRange, importSource, GenericExtractor
packages/graph-core/src/plugins/extractors/go-extractor.ts	extractParams, extractResultType, extractReceiverType, isExported, GoExtractor
packages/graph-core/src/plugins/extractors/java-extractor.ts	extractParams, extractReturnType, hasModifier, extractScopedIdentifierPath, lastComponent, JavaExtractor
packages/graph-core/src/plugins/extractors/php-extractor.ts	extractParams, extractReturnType, extractUseName, lastSegment, PhpExtractor
packages/graph-core/src/plugins/extractors/python-extractor.ts	extractParams, extractReturnType, unwrapDecorated, PythonExtractor

File	Symbols / defs (regex)
packages/graph-core/src/plugins/extractors/ruby-extractor.ts	extractParams, extractAttrProperties, getStringContent, RubyExtractor
packages/graph-core/src/plugins/extractors/rust-extractor.ts	extractParams, extractReturnType, isPublic, extractScopedPath, RustExtractor
packages/graph-core/src/plugins/extractors/types.ts	LanguageExtractor
packages/graph-core/src/plugins/extractors/typescript-extractor.ts	extractParams, extractReturnType, extractImportSpecifiers, TypeScriptExtractor

Module import edges (tabular)

Use together with the flowchart in the Diagrams PDF for a "call graph"-style overview at module boundaries.

From	To
electron/code-city-model.cjs	electron/git-blame.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs
electron/gh-install.cjs	electron/gh-path.cjs
electron/git-commit.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/github-service.cjs
electron/github-auth-flow.cjs	electron/gh-path.cjs
electron/github-commit-context.cjs	electron/github-service.cjs
electron/github-service.cjs	electron/git-service.cjs
electron/github-service.cjs	electron/gh-path.cjs
electron/github-service.cjs	electron/github-commit-context.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/main.cjs	electron/llm.cjs
electron/main.cjs	electron/prefetch-summaries.cjs
electron/main.cjs	electron/novadiff-docs-writer.cjs
electron/main.cjs	electron/novadiff-docs-pdf.cjs
electron/main.cjs	electron/novadiff-docs-html.cjs
electron/main.cjs	electron/file-summary-export.cjs
electron/main.cjs	electron/code-city-model.cjs
electron/main.cjs	electron/knowledge-graph-runner.cjs
electron/main.cjs	electron/security-scan-runner.cjs
electron/main.cjs	electron/git-service.cjs
electron/main.cjs	electron/github-service.cjs
electron/main.cjs	electron/git-commit.cjs
electron/main.cjs	electron/repo-discovery.cjs
electron/main.cjs	electron/git-publish.cjs
electron/main.cjs	electron/workspace-store.cjs
electron/main.cjs	electron/git-blame.cjs
electron/main.cjs	electron/workspace-files.cjs
electron/main.cjs	electron/workspace-history.cjs
electron/main.cjs	electron/github-auth-flow.cjs
electron/main.cjs	electron/gh-path.cjs
electron/main.cjs	electron/gh-install.cjs
electron/main.cjs	electron/export-snapshot.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/prefetch-summaries.cjs	electron/llm.cjs
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs
electron/prefetch-summaries.cjs	electron/file-summary-export.cjs
electron/prefetch-summaries.cjs	electron/git-service.cjs
electron/repo-discovery.cjs	src/app/types.ts
electron/security-scan-runner.cjs	electron/git-service.cjs
electron/workspace-history.cjs	electron/git-service.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
packages/graph-core/src/__tests__/ignore-filter.test.ts	packages/graph-core/src/ignore-filter.ts
packages/graph-core/src/__tests__/ignore-generator.test.ts	packages/graph-core/src/ignore-generator.ts

Cross-file call edges (tabular)

Edges appear only when a call site name uniquely resolves to a single definition path (very conservative).

From file	To file	Callee
cli/src/main.rs	electron/git-service.cjs	push
electron/code-city-model.cjs	electron/git-blame.cjs	blameFileOwnership
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs	normalizeMarkdownDocument
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs	htmlToPdfFile
electron/gh-install.cjs	electron/gh-path.cjs	augmentPathForCli
electron/gh-install.cjs	electron/gh-path.cjs	isGhInstalled
electron/gh-install.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/gh-install.cjs	electron/gh-path.cjs	clearGhCache
electron/git-commit.cjs	electron/git-service.cjs	runGit
electron/git-commit.cjs	electron/git-service.cjs	tryRunGit
electron/git-publish.cjs	electron/git-service.cjs	isGitRepo
electron/git-publish.cjs	electron/git-service.cjs	getRepoStatus
electron/git-publish.cjs	electron/github-service.cjs	getAuthStatus
electron/git-publish.cjs	electron/github-service.cjs	slugFromRepoRoot
electron/git-publish.cjs	electron/git-service.cjs	stagePaths
electron/git-publish.cjs	electron/git-service.cjs	stageAll
electron/git-publish.cjs	electron/git-service.cjs	commit
electron/git-publish.cjs	electron/git-service.cjs	push
electron/git-publish.cjs	electron/github-service.cjs	createPullRequest
electron/github-auth-flow.cjs	electron/gh-path.cjs	isGhInstalled
electron/github-auth-flow.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-auth-flow.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-auth-flow.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-commit-context.cjs	electron/github-service.cjs	getAuthStatus
electron/github-commit-context.cjs	electron/github-service.cjs	tryRunGh
electron/github-service.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-service.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-service.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-service.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/graph/extract-structure.mjs	packages/graph-core/src/plugins/parsers/index.ts	registerAllParsers
electron/knowledge-graph-runner.cjs	packages/graph-core/src/analyzer/graph-builder.ts	GraphBuilder
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveGraph
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveMeta
electron/llm.cjs	packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	process
electron/main.cjs	electron/gh-path.cjs	augmentPathForCli
electron/main.cjs	electron/compare-runner.cjs	runCompareEngineAsync

From file	To file	Callee
electron/main.cjs	electron/compare-runner.cjs	runCompareEngine
electron/main.cjs	electron/llm.cjs	summarizeChange
electron/main.cjs	electron/llm.cjs	summarizeChangeStream
electron/main.cjs	electron/llm.cjs	probeProvider
electron/main.cjs	electron/prefetch-summaries.cjs	stopSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	clearSummaryPrefetchCache
electron/main.cjs	electron/prefetch-summaries.cjs	startSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	getPrefetchedSummary
electron/main.cjs	electron/file-summary-export.cjs	exportFileSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readFileSummaryMarkdowns
electron/main.cjs	electron/file-summary-export.cjs	exportSelectionSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readSelectionSummaryMarkdowns
electron/main.cjs	electron/security-scan-runner.cjs	runSecurityInsightScan
electron/main.cjs	electron/code-city-model.cjs	buildCodeCityModelPayload
electron/main.cjs	electron/novadiff-docs-writer.cjs	writeNovadiffDocsBundle
electron/main.cjs	electron/novadiff-docs-html.cjs	writeNovadiffDocsHtml
electron/main.cjs	electron/novadiff-docs-pdf.cjs	generateNovadiffDocsPdf
electron/main.cjs	electron/novadiff-docs-html.cjs	refreshNovadiffDocsHtmlFromDisk
electron/main.cjs	electron/knowledge-graph-runner.cjs	buildKnowledgeGraph
electron/main.cjs	electron/git-service.cjs	detectGitTooling
electron/main.cjs	electron/github-service.cjs	getAuthStatus
electron/main.cjs	electron/git-service.cjs	getRepoStatus
electron/main.cjs	electron/repo-discovery.cjs	discoverRepos
electron/main.cjs	electron/repo-discovery.cjs	matchRepoForGithubRepo
electron/main.cjs	electron/github-service.cjs	listRepos
electron/main.cjs	electron/github-service.cjs	listPullRequests
electron/main.cjs	electron/git-service.cjs	isGitRepo
electron/main.cjs	electron/git-service.cjs	preparePrCompareRoots
electron/main.cjs	electron/github-service.cjs	viewPullRequest
electron/main.cjs	electron/git-publish.cjs	previewPublish
electron/main.cjs	electron/git-publish.cjs	executePublish
electron/main.cjs	electron/git-service.cjs	stagePaths
electron/main.cjs	electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/main.cjs	electron/git-service.cjs	stageDistrict
electron/main.cjs	electron/workspace-store.cjs	loadSession
electron/main.cjs	electron/github-service.cjs	listDetectedAccounts
electron/main.cjs	electron/github-service.cjs	getUserProfile
electron/main.cjs	electron/gh-install.cjs	getGhToolingStatus
electron/main.cjs	electron/gh-install.cjs	installGh
electron/main.cjs	electron/workspace-store.cjs	setLocalOnlyMode
electron/main.cjs	electron/github-auth-flow.cjs	startGithubDeviceAuth
electron/main.cjs	electron/github-auth-flow.cjs	cancelGithubDeviceAuth
electron/main.cjs	electron/workspace-store.cjs	setGitUser
electron/main.cjs	electron/workspace-store.cjs	workspacesRoot
electron/main.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/main.cjs	electron/workspace-store.cjs	createWorkspace
electron/main.cjs	electron/workspace-history.cjs	indexWorkspaceHistory
electron/main.cjs	electron/workspace-store.cjs	getWorkspace
electron/main.cjs	electron/workspace-store.cjs	setActiveWorkspace
electron/main.cjs	electron/workspace-store.cjs	listWorkspaces
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceLiveRepo
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceUiState
electron/main.cjs	electron/git-blame.cjs	blameFileAtRef
electron/main.cjs	electron/git-commit.cjs	getCommitDetail

Codebase outline (JSON excerpt)

```
{
  "by_extension": [
    {
      "count": 226,
      "ext": ".ts"
    },
    {
      "count": 85,
      "ext": ".tsx"
    },
    {
      "count": 58,
      "ext": ".pak"
    },
    {
      "count": 28,
      "ext": ".cjs"
    },
    {
      "count": 20,
      "ext": "(no ext)"
    },
    {
      "count": 14,
      "ext": ".json"
    },
    {
      "count": 9,
      "ext": ".plist"
    },
    {
      "count": 6,
      "ext": ".css"
    },
    {
      "count": 4,
      "ext": ".dylib"
    },
    {
      "count": 4,
      "ext": ".mjs"
    },
    {
      "count": 4,
      "ext": ".svg"
    },
    {
      "count": 3,
      "ext": ".md"
    },
    {
      "count": 3,
      "ext": ".yaml"
    },
    {
      "count": 2,
      "ext": ".blockmap"
    },
    {
      "count": 2,
      "ext": ".html"
    },
    {
      "count": 2,
      "ext": ".icns"
    },
    {
      "count": 1,
      "ext": ".asar"
    },
    {
      "count": 1,
      "ext": ".bin"
    },
    {
      "count": 1,
```

```

    "ext": ".dat"
  },
  {
    "count": 1,
    "ext": ".dmg"
  },
  {
    "count": 1,
    "ext": ".ico"
  },
  {
    "count": 1,
    "ext": ".jpg"
  },
  {
    "count": 1,
    "ext": ".lock"
  },
  {
    "count": 1,
    "ext": ".nib"
  },
  {
    "count": 1,
    "ext": ".png"
  },
  {
    "count": 1,
    "ext": ".rs"
  },
  {
    "count": 1,
    "ext": ".toml"
  },
  {
    "count": 1,
    "ext": ".txt"
  },
  {
    "count": 1,
    "ext": ".yaml"
  },
  {
    "count": 1,
    "ext": ".zip"
  }
],
"class_diagram_mermaid": "classDiagram\n class CSharpExtractor\n class ChangeAnalysis\n class ChangeKind\n class ClassFingerprint\n class CppExtractor\n class DockerfileParser\n class DroppedEdge\n class EnvParser\n class FileAnalysisMeta\n class FileChangeResult\n class FileFingerprint\n class FileMeta\n class FingerprintStore\n class FrameworkRegistry\n class FunctionFingerprint\n class GenericExtractor\n class GoExtractor\n class GraphBuilder\n class GraphQLParser\n class IgnoreFilter\n class ImportFingerprint\n class JSONConfigParser\n class JavaExtractor\n class LLMFileAnalysis",
"cross_file_call_edges": [
  {
    "from_file": "cli/src/main.rs",
    "to_file": "electron/git-service.cjs",
    "via": "push"
  },
  {
    "from_file": "electron/code-city-model.cjs",
    "to_file": "electron/git-blame.cjs",
    "via": "blameFileOwnership"
  },
  {
    "from_file": "electron/file-summary-export.cjs",
    "to_file": "electron/novadiff-docs-html.cjs",
    "via": "normalizeMarkdownDocument"
  },
  {
    "from_file": "electron/file-summary-export.cjs",
    "to_file": "electron/novadiff-docs-pdf.cjs",
    "via": "htmlToPdfFile"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "augmentPathForCli"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",

```

```

    "via": "isGhInstalled"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "resolveGhExecutable"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "clearGhCache"
  },
  {
    "from_file": "electron/git-commit.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "runGit"
  },
  {
    "from_file": "electron/git-commit.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "tryRunGit"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "isGitRepo"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "getRepoStatus"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/github-service.cjs",
    "via": "getAuthStatus"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/github-service.cjs",
    "via": "slugFromRepoRoot"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "stagePaths"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "stageAll"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "commit"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "push"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/github-service.cjs",
    "via": "createPullRequest"
  },
  {
    "from_file": "electron/github-auth-flow.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "isGhInstalled"
  },
  {
    "from_file": "electron/github-auth-flow.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "ghNotFoundError"
  },
  {
    "from_file": "electron/github-auth-flow.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "resolveGhExecutable"
  },

```



```
{
  "from_file": "electron/github-auth-flow.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-commit-context.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "getAuthStatus"
},
{
  "from_file": "electron/github-commit-context.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "tryRunGh"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "ghNotFoundError"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "parseGithubSlugFromUrl"
},
{
  "from_file": "electron/graph/extract-structure.mjs",
  "to_file": "packages/graph-core/src/plugins/parsers/index.ts",
  "via": "registerAllParsers"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/analyzer/graph-builder.ts",
  "via": "GraphBuilder"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/persistence/index.ts",
  "via": "saveGraph"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/persistence/index.ts",
  "via": "saveMeta"
},
{
  "from_file": "electron/llm.cjs",
  "to_file": "packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts",
  "via": "process"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/compare-runner.cjs",
  "via": "runCompareEngineAsync"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/compare-runner.cjs",
  "via": "runCompareEngine"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/llm.cjs",
  "via": "summarizeChange"
},
{
  "from_file": "electron/main.cjs",
```

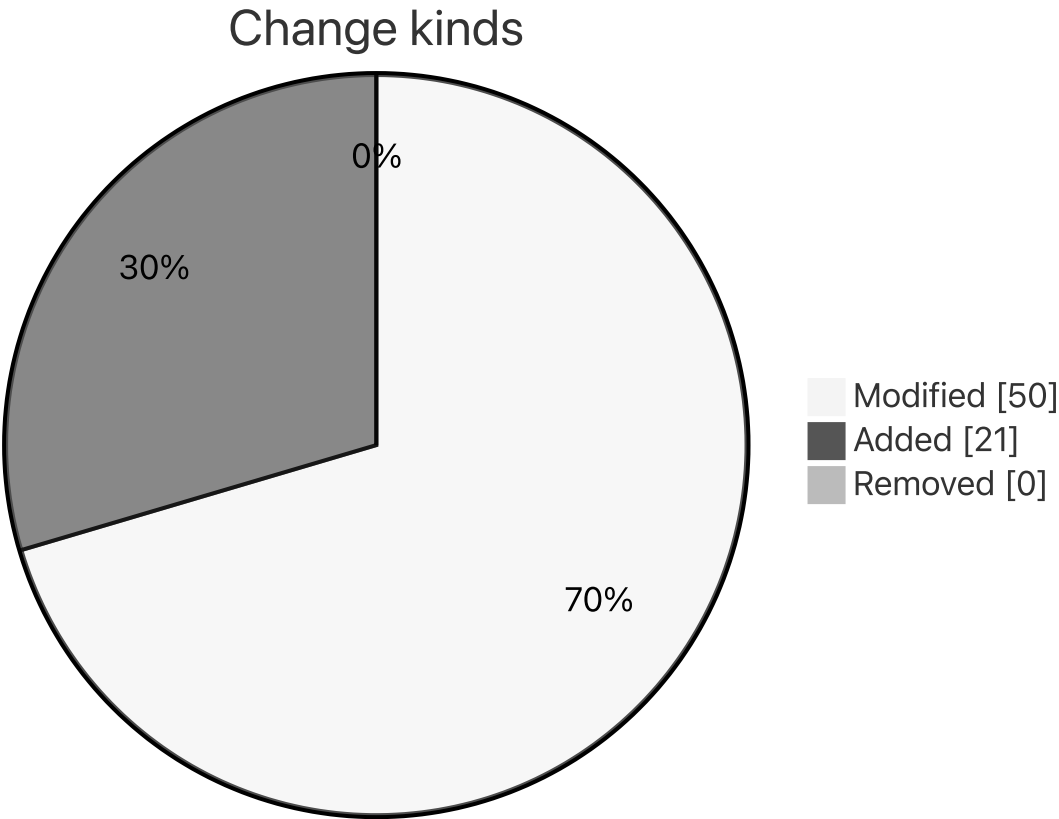
```
"to_file": "electron/llm.cjs",
"via": "summarizeChangeStream"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/llm.cjs",
  "via": "probeProvider"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/prefetch-summaries.cjs",
  "via": "stopSummaryPrefetchWorker"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/prefetch-summaries.cjs",
  "via": "clearSummaryPrefetchCache"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/prefetch-summaries.cjs",
  "via": "startSummaryPrefetchWorker"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/prefetch-summaries.cjs",
  "via": "getPrefetchedSummary"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "exportFileSummaryArtifacts"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "readFileSummaryMarkdowns"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "exportSelectionSummaryArtifacts"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/file-summary-export.cjs",
  "via": "readSelectionSummaryMarkdowns"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/security-scan-runner.cjs",
  "via": "runSecurityInsightScan"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/code-city-model.cjs",
  "via": "buildCodeCityModelPayload"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-writer.cjs",
  "via": "writeNovadiffDocsBundle"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-html.cjs",
  "via": "writeNovadiffDocsHtml"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-pdf.cjs",
  "via": "generateNovadiffDocsPdf"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/novadiff-docs-html.cjs",
  "via": "refreshNovadiffDocsHtmlFromDisk"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/knowledge-graph-runner.cjs",
  "via": "buildKnowledgeGraph"
}
```

```

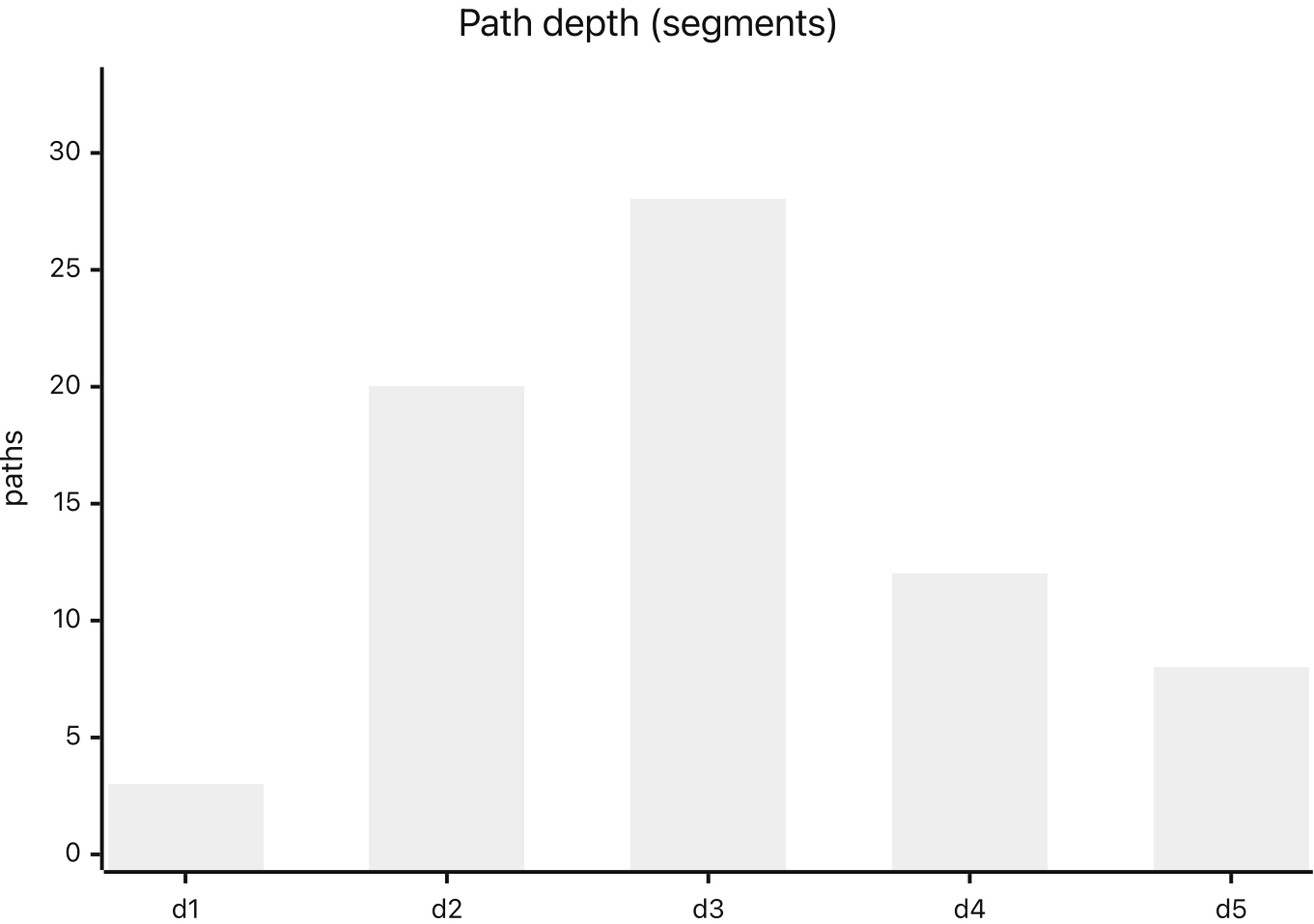
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "detectGitTooling"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "getAuthStatus"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "getRepoStatus"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/repo-discovery.cjs",
  "via": "discoverRepos"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/repo-discovery.cjs",
  "via": "matchRepoForGithubRepo"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "listRepos"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "listPullRequests"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "isGitRepo"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "preparePrCompareRoots"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "viewPullRequest"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-publish.cjs",
  "via": "previewPublish"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-publish.cjs",
  "via": "executePublish"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "stagePaths"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/export-snapshot.cjs",
  "via": "exportProjectSnapshotZip"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "stageDistrict"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/workspace-store
... truncated ...

```

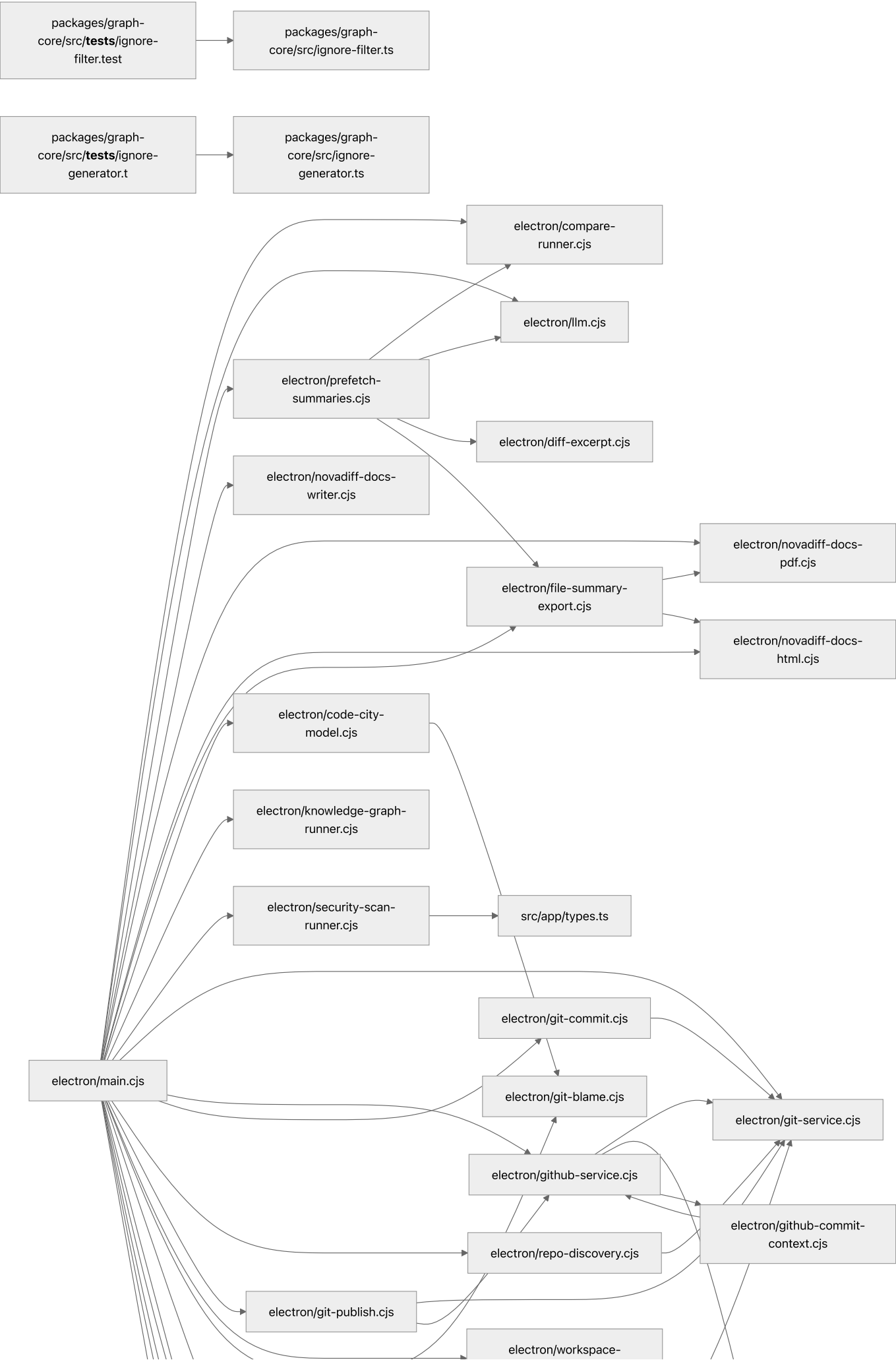
Change mix

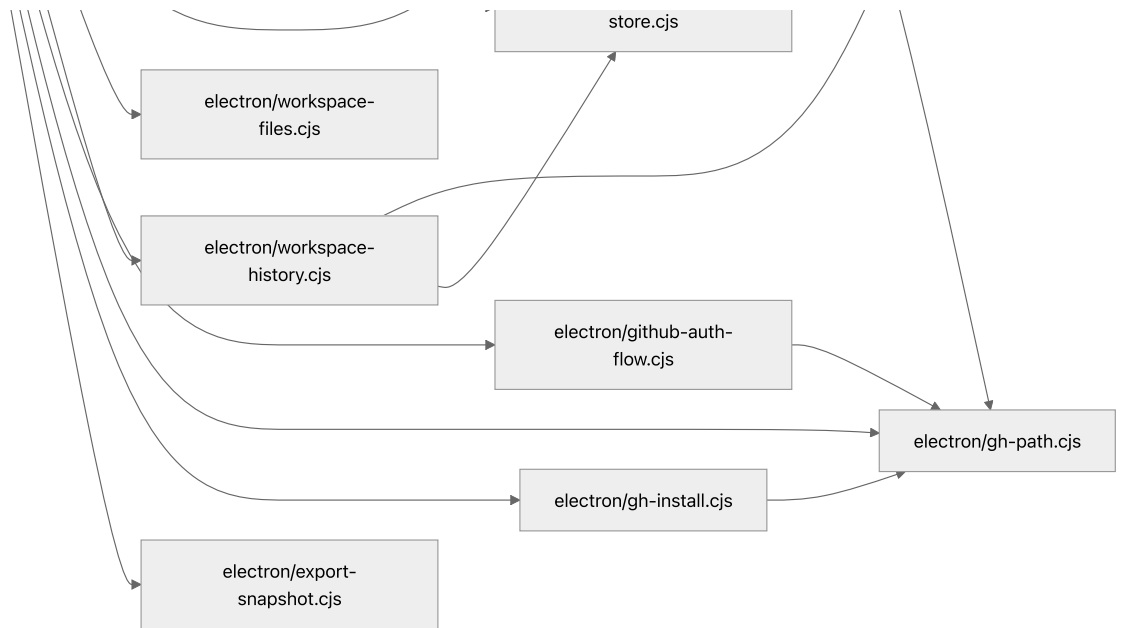


Path depth



Module import graph (relative imports)





Cross-file calls (heuristic)

